

Flight Test Planning Document

Experimental activities at Pipistrel

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List of symbols, abbreviations and acronyms

Acronym	Definition
AGL	Above Ground Level
AHRS	Artificial Horizon System
ALT	Altimeter
AOA	Angle of Attack
ASI	Airspeed Indicator
ASL	Above Sea Level
ATZ	Air Traffic Aerodrome
CAS	Calibrated Airspeed
CTR	Control Zone
EGT	Exhaust Gas Temperature
FTP	Flight Test Pilot
FTE	Flight Test Engineer
FTI	Flight Testing Instrumentation
GCS	Ground Control System
HSI	Horizontal Situation Indicator
IAS	Indicated Airspeed
MAP	Manifold Absolute Pressure
OAT	Outside Air Temperature
PFD	Primary Flight Display
RPM	Revolutions Per Minute
SOC	State of Charge
SOH	State of Health
STOL	Short Takeoff and Landing
TAS	True Airspeed
VHF	Very High Frequency
VOR	Visual Flight Rules
WAAS	Wide Area Augmentation System

Symbol	Definition
V_{NE}	Never Exceed Speed
V_{FE}	Maximum Flap Extended Speed
V_{AE}	Max Full Airbrakes Speed
V_s	Stall speed - clean (flaps 0)
V_{s1}	Stall speed - take-off (flaps +1)
V_{s0}	Stall speed - landing (flaps +2)
V_{trim}	Trim speed
V_R	Rotation Speed
V_{LOF}	Lift-Off Speed
V_{50}	Airspeed at 50 ft AGL
V_{TD}	Touchdown Speed
S_g	Ground Roll Distance
S_{TO}	Total Take-Off Distance
S_3	Distance: obstacle - touchdown
S_4	Distance: touchdown - stop
S_{LND}	Total Landing Distance
$MTOW / MLW$	Max Takeoff / Landing Weight
M_F	Total usable fuel
R	Range (LND @ 30% SOC)
E	Endurance (LND @ 30% SOC)

Abbreviation	Definition
TO	Takeoff
LND	Landing
GLD	Glider

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1 Introduction

1.1 General objectives

The objective of the proposed flight test planning document is to verify and validate the performance, handling qualities, and operational characteristics of the Pipistrel SW121 Explorer and SW128 Electro aircraft, in accordance with the established test plan. The trials are designed to gather both quantitative and qualitative data in order to assess conformity with design expectations, applicable regulatory requirements, and relevant safety standards. To ensure consistency and comparability, each mission will be executed following an identical flight profile on both aircraft. The test programme further seeks to identify any operational or performance differences between the two variants, while confirming system functionality and compliance with certification criteria. All activities shall be carried out in adherence to stringent safety procedures and within the defined operational limitations of each aircraft. A key challenge in the planning phase lies in designing a comparable mission profile for both test aircraft, while ensuring compliance with their respective performance limitations and endurance constraints. This process requires careful balancing of technical requirements with operational feasibility. Moreover, the planning environment presents a new context in which a highly objective-driven and practically oriented approach must be adopted. It necessitates the integration of diverse engineering competencies to ensure, above all, **operational safety**, while also aiming to optimise efficiency in terms of both time and cost across the test campaign.

1.2 Test aircraft

Both employed aircraft are classified as ultralight and are manufactured by Pipistrel, but they differ significantly in propulsion system and performance characteristics, aspects relevant to the specific tests to be conducted.

Explorer – SW121

The Pipistrel SW 121 Explorer is a high-wing, T-tail, two-seat ultralight aircraft equipped with a fixed tricycle landing gear and powered by a front-mounted Rotax 912 ULS engine producing 100 hp. It features a constant-speed (variable-pitch) propeller and is certified in the CS-LSA category.

The aircraft is not designed for aerobatic flight. Only those manoeuvres incidental to normal operations are approved, including normal stalls, chandelles, lazy eights, and turns with bank angles not exceeding 60°. Intentional spinning is permitted only under specific conditions: with flaps set to 0, airbrakes retracted, and engine power at idle. The Explorer has a MTOW of 600 kg. It is a versatile platform suited to a variety of operational tasks, including training, touring, and utility roles.

The following table summarises the main geometric and operational characteristics of the aircraft.

Basic Dimensions	Value
Length	6.42 m
Span	10.70 m
Height *	2.08 m
Wing	Value
Area	9.51 m ²
Span	10.70 m
Mean aerodynamic chord	0.898 m

Horizontal Tail	Value
Area	1.39 m ²
Span	2.18 m
Vertical Tail	Value
Area	1.24 m ²
Height	1.11 m
Weights	Value
Maximum takeoff weight	600 kg
Typical empty weight	371 kg
Typical useful load	229 kg
Maximum baggage weight	25 kg

TABLE 1: EXPLORER – SW121 GENERAL INFORMATION.

Velis Electro – SW128:

The Pipistrel Velis Electro is a high-wing, T-tail, two-seat aircraft equipped with a fixed tricycle landing gear and a front-mounted, liquid-cooled electric motor. It is fitted with a fixed-pitch propeller and is certified for *VFR day* operations under EASA CS-LSA regulations. The aircraft is explicitly intended for *non-aerobatic* use, which includes only those manoeuvres incidental to normal flight, such as stalls (excluding whip stalls), figure-eights, chandelles, and steep turns not exceeding 60° of bank. All spins are strictly prohibited.

The Velis Electro delivers a nominal power output of 57.6 kW. Two different battery configurations are available: PB345V124E-L and PB345V119E-L. With a MTOW of 600 kg. The table below presents the principal geometric and operational characteristics of the aircraft.

Basic Dimensions	Value
Length	6.47 m
Span	10.71 m
Height	2.06 m
Wing	Value
Area	9.51 m ²
Span	10.71 m
Mean aerodynamic chord	0.898 m
Horizontal Tail	Value
Area	1.39 m ²
Span	2.18 m
Vertical Tail	Value
Reference area	1.24 m ²
Height	1.11 m
Weights	Value
Maximum take off weight	600 kg
Design empty weight	428 kg *
Design useful load	172 kg *
Maximum baggage weight	No baggage

TABLE 2: ELECTRO – SW128 GENERAL INFORMATION.

1.3 Test instrumentation

The following tables gather all the testing instrumentation present onboard for both the SW121 and SW128. The instrumentation is divided between **native** and **additional**.

Explorer - SW121

Sensor	Function	Measured Data
Garmin GEA 24 EIS	Engine Interface System	Coolant temperature, Exhaust Gas Temperature (EGT), Manifold Absolute Pressure (MAP), oil pressure, oil temperature, fuel pressure, fuel flow, voltage, current (amperage), engine RPM. Sampling frequency: 10Hz .
Garmin GSU 25C ADAHRS	Air Data, Attitude and Heading Reference System	Roll, pitch, yaw, magnetic heading, tri-axial acceleration, Outside Air Temperature (OAT), True Airspeed (TAS) and wind vector calculations. HSI (horizontal situation indicator). Utilises data from a remote magnetometer for enhanced accuracy. Sampling frequency: 10Hz .
Garmin GAP 26	Pitot-Static System with AoA Sensing	Total and static pressure, indicated airspeed, angle of attack. Sampling frequency: 10Hz .
Mechanical Altimeter	Mechanical Altitude Measurement	Three-pointer barometric altimeter, independent of electrical power.
Garmin GMU 22	Magnetometer	Measures the Earth's magnetic field to determine magnetic heading; operates in conjunction with the ADAHRS. Sampling frequency: 10Hz .
Magnetic Compass	Liquid Magnetic Compass	Indicates magnetic heading. Serves as an independent backup navigation aid.
Slip Indicator	Inclinometer	Indicates turn coordination using a free-moving ball in a curved, oil-filled tube (passive instrument).
Turn Coordinator	Turn Rate and Coordination	Displays rate of turn and coordination; function integrated into the Primary Flight Display (PFD).
Garmin GNC 355A	COM + WAAS GPS	VHF communications and GPS navigation with Wide Area Augmentation System (WAAS); supports RNAV procedures. Sampling frequency: 1Hz .
Garmin GNC 255A	COM/NAV Radio	Provides VHF communication and receives navigation signals from VOR, Localiser (LOC), and Glideslope. Supports 8.33 kHz channel spacing.
Garmin GTX 345	ADS-B Transponder	Transmits aircraft position, altitude and velocity via ADS-B Out; receives traffic (TIS-B) and weather data (FIS-B) via ADS-B In.

TABLE 3: EXPLORER - SW121 NATIVE INSTRUMENTATION.

Sensor	Measured Data
FTI Controller + Display	Displays real-time flight test data acquired from onboard sensors. Event marking and Windowing (DATA BLOCK or TOP).
Potentiometer	Records angular displacement of elevator control surface.

GPS + RTK	Provides high-precision ground speed, altitude, and velocity vector components; also enables real-time telemetry transmission to the Ground Control Station (GCS).
Cockpit Camera + Voice	Recording of the mission of both audio and video from the plane cockpit.

TABLE 4: EXPLORER - SW121 ADDITIONAL INSTRUMENTATION.

Electro - SW128

Sensor	Function	Measured Data
EPSI570C	Integrated Avionics	Motor operating status, RPM, power and temperature. Power controller operating status, temperature. Battery system operating status, temperature, SOC and SOH. Auxiliary (14V) battery status and voltage. Sampling frequency: 16Hz .
Kanardia Horis (AH)	Artificial Horizon	Provides aircraft attitude: pitch and roll, using internal sensors and LCD display. OAT measurement. Sampling frequency: 16Hz .
Garmin Aera 660	GPS Navigator	Provides GNSS-based position, navigation, and moving map display. Sampling frequency: 1Hz .
Kanardia VSI	Vertical Speed Indicator	Measures rate of climb or descent; displayed with needle and digital readout. Sampling frequency: 16Hz .
Mechanical Altimeter	Barometric Altimeter	Indicates altitude based on local barometric pressure; conventional three-pointer.
Mechanical ASI	Airspeed Indicator	Measures indicated airspeed via Pitot-static pressure difference (knots).
Funke ATR833 COM	Radio Communication	Provides VHF voice communication; supports 8.33 / 25 kHz channel spacing.
Funke TRT800H	Mode S Transponder	Transmits aircraft identification, position, and altitude; includes altitude encoder.
Mechanical Compass	Magnetic Heading	Displays aircraft heading relative to magnetic north; liquid-filled, manual.
Slipball Winter	Turn and Slip Indicator	Indicates turn rate and coordination using an inclinometer and gyroscopic input.

TABLE 5: ELECTRO - SW128 NATIVE INSTRUMENTATION.

Sensor	Measured Data
FTI Controller + Display	Displays real-time flight test data acquired from onboard sensors. Event marking and Windowing (DATA BLOCK or TOP).
Potentiometer	Records angular displacement of elevator control surface.
GPS + RTK	Provides high-precision ground speed, altitude, and velocity vector components; also enables real-time telemetry transmission to the Ground Control Station (GCS).
Cockpit Camera + Voice	Recording of the mission of both audio and video from the plane cockpit.

TABLE 6: ELECTRO - SW128 ADDITIONAL INSTRUMENTATION.

Additional Instruments for Mission Reconstruction and Data Analysis

To facilitate accurate mission reconstruction and enhance post-flight data analysis, the following additional instruments will be employed during both the SW121 and SW128 test flights:

- **Chronometer:** Used to log precise time references for events and test procedures, allowing synchronisation with recorded telemetry and instrument readings.
- **Control fixer:** The control fixer is a device, typically a rubber band, used during stick-fixed phugoid tests. It is installed on the FTP's leg once the aircraft is properly trimmed and serves as a physical reference for the control stick's equilibrium position. After the singlet input is applied to initiate the phugoid motion, the pilot shall return the longitudinal control to the position indicated by the control fixer and maintain it there. This ensures that the stick remains in the trimmed (neutral) position, avoiding any unintended elevator inputs. The device provides the pilot with a stable, tactile reference point to preserve control fidelity throughout the manoeuvre.

1.4 Test range

Airport Description

The flight test activities will be conducted at Gorizia Airport "Amedeo Duca d'Aosta", officially designated with ICAO code **LIPG** and IATA code **QGO**. The airport is located approximately 4 km southwest of the city of Gorizia, near the border with Slovenia, with geographical coordinates **45°54'24"N, 13°35'57"E**, and an elevation of **207 feet** (63 meters) above sea level. The airfield spans approximately 150 hectares and features two grass runways. The primary runway (09/27) is **1100 meters long and 60 meters wide**, while the secondary runway (04/22), currently **closed**, measures 890 meters in length. Circuit altitude is **700 feet AGL**, with a left-hand traffic pattern. Gorizia Airport operates in uncontrolled airspace (**Class G**), supporting only VFR (Visual Flight Rules) traffic. No ATS (Air Traffic Services) are provided on-site. Pilots are solely responsible for traffic separation both in the air and on the ground. **Communication** is carried out via self-announced position reports on frequency **119.855 MHz**, which is also monitored by the fire rescue unit during flight operations.

Meteorological conditions are generally characterized by variable light winds, often influenced by seasonal breezes. In winter, east to northeast winds are more common, occasionally reaching gusts up to 10 knots.

Nearby navigational aids include:

- RCH (Ronchi dei Legionari VOR) – 114.20 MHz, bearing 043°, 6.8 NM away
- RIV (Rivolto VOR) – 110.00 MHz, bearing 100°, 22.0 NM
- POR (Portorož VOR) – 115.15 MHz, bearing 354°, 25.9 NM
- ILB (Bistrica VOR) – 114.80 MHz, bearing 306°, 31.6 NM

Airport Procedures

All flight operations at Gorizia Airport must be conducted under visual flight rules (VFR), strictly adhering to the published traffic circuit procedures. Self-coordination among pilots on the traffic frequency is mandatory, as no air traffic control service is provided at the aerodrome. The airport falls under the jurisdiction of *ENAC – Direzione Aeroportuale Nord-Est* – and is subject to applicable regulations, circulars, and ordinances as defined in the Italian Navigation Code.

The standard traffic pattern is flown at **1000 ft above ground level (AGL)**, positioned to the north and north-northwest of the runways. Particular attention must be given to the restricted area LI P76 "Gorizia," within which flight operations are strictly prohibited. This area, updated under AIRAC A8/24 effective from 8 August 2024, has lateral and vertical limits extending from **the surface up to 1500 ft AGL**. For any special activity

requests within this zone, prior clearance must be obtained through the DAP, with reference to AIP section ENR 5.1 para. 7.3 [10].

Due to the proximity of the **Slovenian border**, specific departure and arrival procedures are enforced. Takeoffs are permitted from both runway 27 and 09. After takeoff from runway 27, aircraft must either continue straight on course or execute a right turn. When departing from runway 09, a left turn is required immediately after reaching 706 ft AGL. Landing operations are restricted: **landings are not authorised on runway 27** and may only be conducted on runway 09. The prescribed traffic circuits and associated procedures are illustrated in the figure provided in reference [1].

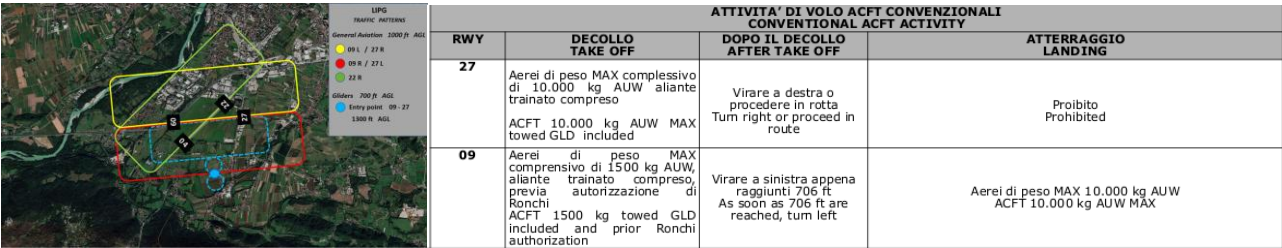


FIGURE 1: TO AND LND PROCEDURES.

Expected Weather Conditions

On May 20th, Gorizia experiences spring-like conditions characterized by moderate temperatures, variable winds, and moderate humidity. The average temperature during this period is approximately 17.7°C, with daytime highs reaching up to 23.3°C and nighttime lows around 12.1°C . Wind patterns in May are typically light, with average speeds around 7.5 km/h. Early in the month, winds predominantly come from the east, transitioning to southerly directions later in the month. Relative humidity levels average around 64% to 70%, contributing to a comfortable atmosphere for flight operations [11].

Expected Test Location and Mission Overview

The flight testing will begin at Gorizia Airport with take-off from runway 27, followed by climb-out and departure from the ATZ towards a pressure altitude of 3'000 ft. Stick-free phugoid tests will be flown northwest toward the Manzano area. A square pattern near Manzano will be used for wind assessment and stall testing. The return leg heads southeast back toward Gorizia, where stick-fixed phugoid tests will be conducted before entering the circuit for landing tests.

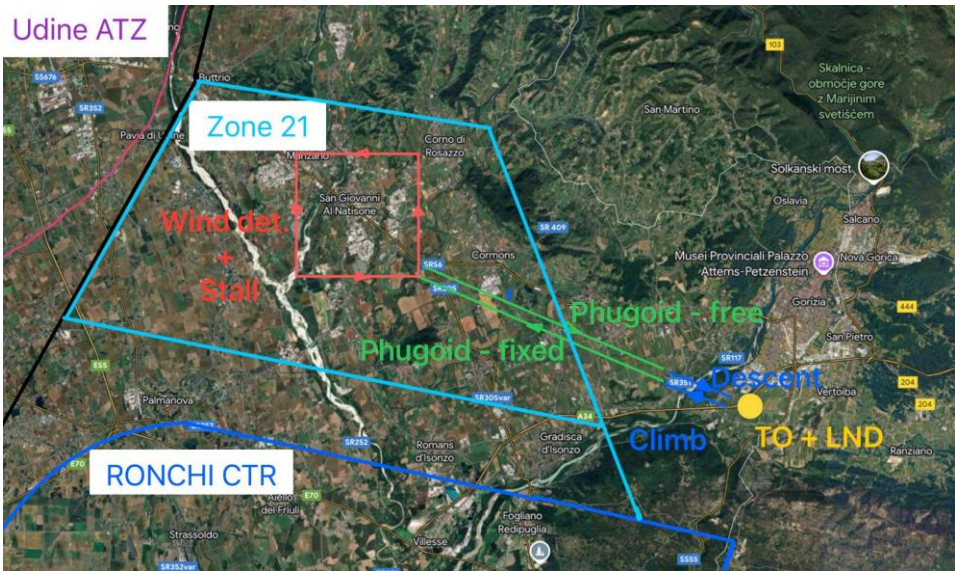


FIGURE 2: TEST RANGE AND PLANNED TEST LOCATION.

1.5 Limitations

While planning a test mission it is important to plan respecting and keeping into consideration all the several limitations and restriction that, the weather, airspace, borders, geographical obstacles and aircraft limitations. Following there are reported the main limitations that are relevant for the design and planning of the test mission.

Airspace

The airspace surrounding Gorizia Airport presents several important operational considerations. The airport is located within an Aerodrome Traffic Zone (ATZ), requiring all pilots to communicate their intentions when entering or exiting the zone via the designated radio frequency. Compliance with the specific operational procedures of the LIPG ATZ is mandatory and detailed in Section 1.4 – *Airport Description and Procedures*. In addition, a prohibited airspace—designated P76—extends above the centre of Gorizia. Flight within this area is strictly forbidden from the surface up to 1,500 ft AGL.

Two other controlled airspace zones are also relevant to this mission and must be considered during planning and operations:

- **Zone 21 “PALMANOVA”**: imposes a vertical limitation, the airspace is divided into 2 sections vertically:
 - SFC / 3'000 ft AMSL: non-directly-regulated airspace
 - 3'000 ft AMSL / FL85: Regulated airspace, the pilot needs to contact the control entity.
- **RONCHI CTR**: defined vertically from SFC to 1'500 ft AMSL – entering and exiting this airspace is controlled by the RONCHI TMA.
- **Udine ATZ**: entering and exiting this ATZ needs to be regulated contacting Udine Airport at the radio frequency **119.055MHz** [11].
- **Slovenian Border**: in order to cross the border, the flight would need to be planned in advance and the Slovenian entity would need to be warned.

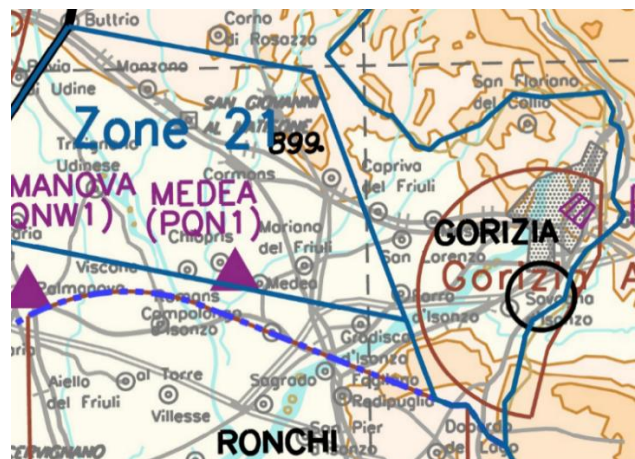


FIGURE 3: MISSION AIRSPACE LIMITATIONS.

Takeoff and Landing

When performing the landing and takeoff manoeuvres, as previously mentioned in section 1.4, there are some important limitations that are worth repeating.

- **Takeoff allowed from both runway 27 and 09** (only for aircrafts with MTOW under 1'500 kg).
- **Landing prohibited from runway 09**. Landing is exclusively allowed in runway 27.

Additionally, every entering and exiting the Gorizia ATZ must be communicated.

Aircraft (POH)

The primary constraints to be considered for this mission are those related to the aircraft’s operational capabilities. The entire mission must be both planned and executed well within these limitations, with **flight safety being the top priority at all times**. The main aircraft limitations that have influenced the mission planning are summarised in the following tables, along with an explanation of how each has been taken into account. Each table is specific for either the SW121 – Explorer or the SW128 – Velis Electro aircraft.

Note: CG position and aircraft weight are not yet defined, since they are dependent on the fuel load and weight of the crew on the testing day, therefore it remains a variable of this planning.

Symbol	Definition	Value	Mission Impact
V_{NE}	Never Exceed Speed. Calibrated to 3'000 ft DA.	156 KIAS	Stall recovery dive and phugoid peak airspeed.
V_{FE}	Max Flap Extended Speed	+1: 81 KIAS +2: 65 KIAS 0: 135 KIAS	Phugoid airspeed peaks. Takeoff and landing testing procedures.
V_{AE}	Max Full Airbrakes Speed	100 KIAS	Descent/approach procedure.
V_{S0}	Stall Speed (+2)	47 KIAS	Landing testing.
V_{s1}	Stall Speed (+1)	51 KIAS	Takeoff testing.
V_s	Stall Speed (0)	53 KIAS	Phugoid safe margins and stall testing.
V_{cw}	Max Crosswind Speed	18 kn	Landing and Takeoff testing.
n	Structural g-load limits	+4.0 g / -2.0 g	Safe manoeuvring.
$Max\ h_{TO}$	Maximum takeoff field elevation	10'000 ft AMSL	Restricts choice of departure airfields.
$Max\ h$	Maximum allowed cruise altitude	18'000 ft AMSL	Defines the operational flight ceiling.
$Max\ OAT$	Maximum permissible ambient temperature	+55 °C	Impacts engine performance and airframe resistance.
$MTOW\ /\ MLW$	Max Takeoff / Landing Weight	600 kg	Limits total loading and usable fuel.
Max Seat Weight	Max per seat / total occupant weight	110 kg / 200 kg	Ensure passenger compliance.
M_F	Total usable fuel	100 liters	Range and endurance.
E	Maximum Endurance	3.5 h (3'000 ft)	Mission Total Endurance.
CG @ min TOW	CG limits, min weight (427 kg)	25% MAC / 35% MAC	Pre-flight operations and safety, impact on takeoff, stall, phugoid and land testing.
CG @ max TOW	CG limits, max weight (600 kg)	27.5% MAC / 35% MAC	Pre-flight operations and safety, impact on takeoff, stall, phugoid and land testing.

TABLE 7: SW121 – EXPLORER LIMITATIONS.

Symbol	Definition	Value	Mission Impact
V_{NE}	Never Exceed Speed	108 KIAS	Stall recovery dive and phugoid peak airspeed.
V_{FE}	Max Flap Extended Speed	+1: 81 KIAS +2: 65 KIAS	Phugoid airspeed peaks. Takeoff and landing testing procedures.
V_{S0}	Stall Speed (+2)	46 KIAS	Landing testing.
V_{s1}	Stall Speed (+1)	46 KIAS	Takeoff testing.

V_s	Stall Speed (0)	53 KIAS	Phugoid safe margins and stall testing.
V_{cw}	Max Crosswind Speed	18 kn	Landing and Takeoff testing.
n	Structural g-load limits	+4.0 g / -2.0 g	Key for safe manoeuvres during testing and dynamic events.
Max h	Maximum allowed cruise altitude	13'000 ft AMSL	Defines the operational flight ceiling.
Max OAT	Maximum permissible ambient temperature	+35 °C	Impacts battery performance and planning in hot conditions.
Min OAT	Minimum permissible ambient temperature	-0 °C	Impacts battery performance and planning in cold conditions.
MTOW / MLW	Max Takeoff / Landing Weight	600 kg	Limits total loading and usable fuel.
Max Seat Weight	Max per seat / total occupant weight	110 kg / 200 kg	Ensure passenger compliance.
E	Maximum Endurance (LND with 30% SOC)	45 min (3'000 ft)	Mission Total Endurance.
fwd / aft CG	CG limitations with crew	25.2% MAC / 32.6% MAC	Pre-flight operations and safety, impact on takeoff, stall, phugoid and land testing.

TABLE 8: SW128 – ELECTRO LIMITATIONS.

1.6 Schedule

The flight test mission is expected to follow the sequence outlined below. While this plan represents the optimal flow, minor adjustments may be made in response to real-time weather conditions.

1. Familiarization with FTI.
2. Brief the test pilot and company flight test engineer, confirming the planning.
3. Completion of all pre-flight checklists and begin data acquisition.
4. Taxi to runway 27 and prepare for the TO.
5. Takeoff from runway 27 at Gorizia Airport.
6. Conduct takeoff performance testing.
7. Exit the airport traffic pattern, climb en route, and leave the ATZ of Gorizia. Continue climbing to a pressure altitude of 3,000 ft.

Note: the takeoff, climb and routing steps may vary depending on meteorological conditions.

6. Set a heading of approximately 300°.
7. Perform two phugoid test points (stick free), flying straight for a distance of approximately 8 km.
8. Upon reaching the vicinity of Manzano and San Giovanni Al Natisone, begin flying a square pattern at 3,000 ft.
9. **Remark:** the square pattern consists of four legs, each approximately 4 km long.
10. Complete one full circuit to determine wind conditions for the stall testing.
11. Conduct stall testing during the same flight pattern. A minimum of six test points should be performed over at least two laps.
12. Allocate an additional two laps for repeating or recovering any failed or incomplete test points.

-
13. Turn back toward Gorizia Airport while maintaining altitude at 3,000 ft and a heading of approximately 120°.
 14. Perform two phugoid test points (stick fixed), again covering a straight flight path of around 8 km.

15. As the aircraft approaches the vicinity of the airfield, enter the traffic pattern and prepare for the landing test sequence.
16. Conduct landing performance testing according to the prescribed procedure.
17. Perform a full-stop landing and record the final set of test data.
18. Complete all post-flight checklists.

The figure below describes the planned test mission profile, with the various test objectives labelled.

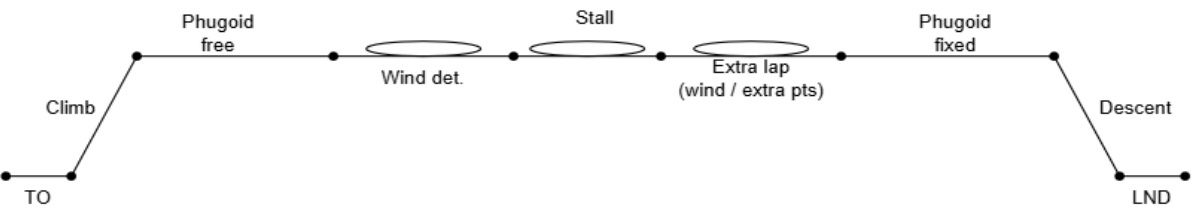


FIGURE 4: TEST MISSION PROFILE.

Finally, an assessment of the total flight time and energy consumption is essential when planning the mission, as it ensures both feasibility and operational soundness. Based on the mission sequence described above, conservative estimates have been made for the fuel consumption of the conventionally powered SW121 and the battery SOC of the electrically powered SW128. These estimations allow for adequate reserves, accounting for potential rerouting, holding, or traffic delays.

All calculations have been conducted using average performance values for each phase of flight, as specified in the respective POHs [5][6]. Furthermore, a conservative battery SOH of 70% has been assumed for the SW128 battery pack.

A summary of the overall energy estimates is provided in the table below:

Aircraft	Total Time [min]	Estimated fuel / battery consumption
SW121	45	17 L
SW128	45	70 %

TABLE 9: MISSION DURATION AND ENERGY CONSUMPTION.

2 Test and evaluation

2.1 General

The procedures for a series of flight tests—including **take-off, stall, phugoid, and landing** evaluations—are outlined in detail in the following sections. Each section provides a structured explanation of the test concept, the minimum required instrumentation necessary to enable effective data analysis during post-processing, and a comprehensive breakdown of the execution plan. Additionally, the data to be recorded by the onboard test engineer will be specified through a standardised flight test card. A dedicated subsection also addresses whether any data reduction, correction, or standardisation procedures, such as compensation for wind effects or weight variations, is present. To support the reliability of data collection, a checklist has been included for each test. This checklist, divided between the responsibilities of the test engineer and the pilot, is intended to **minimise the risk of data loss or measurement failure**. Given the dynamic nature of the testing environment and the limited time available to carry out each procedure, the use of checklists is essential and must be reviewed in advance by both the company's engineering team and the pilot.

Prior to the commencement of any flight testing or any flight, for that matter, the pre-flight checklists must be completed. Presented below is a proposed pre-flight checklist to be carried out by the FTE to ensure that all instrumentation is correctly configured and functioning as intended.

Item	Notes
Pilot and Passenger weight	Record.
TO weight	Record.
TO balance	Record.
Special notes	Record special particularities of the plane / loading / anything.
Engine ON	Pilot turns on aircraft.
SD card – FC log ON	Correctly read from Flight Controller (FC) and logging.
Engine display ON	Set Engine display to show engine info (RPM, MAP/Power) and estimated weight (SW121).
AHRS display ON	Correctly visually reading ASI, PA and OAT. (Simulate with FTI display in front)
FTI log ON	Correctly logging and displaying of the FTI.
FTI windowing OFF	Make sure that FTI windowing is turned OFF .
FTI display ON	Correct FTI display page is set up. UTC must be shown
FTI controller ON	Correctly functioning. Event marker a 0.
Camera ON	Check.
Audio ON	Check.
Pen working and spare	Check.

TABLE 10: FTI PRE-FLIGHT CHECKLIST.

REMARK: The centre of gravity (CG) loading condition has not yet been defined; however, it will ideally be set to an **intermediate position**. This is because stall and take-off testing require diametrically opposed CG positions, while both the Explorer-SW121 and Electro-SW128 will exhibit negligible CG excursion during flight and testing in different CG configurations would mean perform multiple test flight. Following the target CG position as % MAC for the testing flights. **Target SW121 CG: 30.5 %. Target SW128 CG: 30 %.**

2.2 Take-off performance

General Objective

The principal objective of the take-off test is to evaluate the **minimum take-off distance** required by the aircraft when operating under standard conditions. The test procedure will be carried out in accordance with current airfield regulations and will follow the methodologies outlined in *paragraphs 23.53 and 23.59 of CS-23 Amendment 4* [4], as well as *paragraph 4.4.2 of ASTM F2245 – 12d* [2].

These regulatory references define not only the methods for determining take-off performance, but also the appropriate correction procedures to account for environmental and operational factors, such as wind conditions and runway slope. It is acknowledged that take-off performance data are inherently subject to considerable variability, owing to both atmospheric influences and the highly dynamic nature of the take-off phase. Accordingly, standard flight test practice — as reflected in certification guidance — recommends that the test be repeated at least six times in order to support statistically reliable analysis and reduce uncertainty through data regression [4].

However, given the time constraints associated with the current test campaign, and in light of the broader experimental focus on additional aircraft characteristics, the take-off test will be conducted **once only**. Despite the limited number of repetitions, the procedure will be carried out with strict adherence to regulatory guidance, ensuring the operational repeatability and consistency required for valid performance assessment.

Certification Requirements / Tolerances

In order to comply with the relevant certification standards from [2], the following performance conditions must be demonstrated during testing:

- The rotation speed V_R must not be lower than 1.1 times the stall speed in the take-off configuration ($V_R \geq 1.1 \cdot V_{S1}$)
- The aircraft must be in standard take-off configuration (flaps position +1 unless otherwise specified).
- The speed at 50 ft (V_{50}) must exceed 1.3 times V_{S1} from [2] and must be sufficient to achieve both the fast climb and steep climb gradients required by CS-23 paragraph 23.53 [4]. ($V_{50} \geq 1.3 \cdot V_{S1}$)
- Changes in weight inside the $\pm 0.5\%$ the reference test weight are considered acceptable, without the need for a weight correction.
- CG in most unfavourable position: 27.5% MAC for the SW121 and 25.2% MAC for the SW128 (to be determined).

This initial section establishes the framework for the subsequent detailed methodology, test point selection, data acquisition techniques, and analysis procedures that will be implemented throughout the take-off performance evaluation.

Measurements and Data Acquisition

Given the highly dynamic nature of the take-off manoeuvre, the number of measured parameters has been deliberately kept to a minimum in order to reduce complexity and minimise the workload. Certain engine parameters, such as MAP, are treated as secondary variables. This is justified by the assumption that air density remains nearly constant during the brief take-off run and that the throttle setting will be maintained at full power throughout, resulting in relatively stable engine performance during this phase. Nonetheless, in order to enable accurate post-flight data analysis and environmental correction, a set of critical **pre-flight ground measurements** must be obtained. These include: OAT, QNH, QFE, runway slope or inclination, wind speed, wind direction and takeoff weight or total fuel remaining. All other dynamic flight parameters will be recorded through on-board instrumentation and FTI systems. Where applicable, redundancy between

manual and digital logging is maintained to ensure reliability and to enable cross-validation in the event of instrument failure or data anomalies. It is worth noting that, to increase redundancy, the native GPS instrumentation system data will still be recorded from the sensors GNC 355 for the SW121 and the Aera 660 for the SW128 as ulterior backup.

Measurement	Type	Flight Card	Data - SW121	Data - SW128
Time	Required	Chronometer and Event Marker	FTI GPS time	FTI GPS time
Speed	Required	ASI or displayed GPS speed on FT screen	FTI GPS speed	FTI GPS speed
Engine RPM	Secondary	RPM noted from engine display	GEA 24 EIS	EPSI570C
Engine MAP	Secondary	MAP noted from engine display	GEA 24 EIS	EPSI570C
Height	Secondary	From ALT, both QNH and QFE	GSU 25C + GPS FTI	AHRS + GPS FTI
Distance	Secondary	None	FTI GPS speed	FTI GPS distance

TABLE 11: TO MEASUREMENTS.

Test Method

The take-off test shall be conducted in the aircraft's standard take-off configuration (+1), as defined in the applicable sections of the Pilot's Operating Handbook (POH). Prior to initiating the test sequence, it is essential to acquire all relevant runway, atmospheric and aircraft conditions, as these will be critical for data correction and repeatability. The takeoff methodology to be followed is explained as it follows:

1. **Runway Alignment:** The aircraft shall be positioned abeam a measured ground reference marker, such as a runway distance-to-go indicator to establish a consistent starting point. Set altimeter to **QFE regulation** for correctly visualizing the 50 ft condition.
2. **Condition Logging:** Environmental and configuration parameters shall be recorded at this stage: QNH and QFE, OAT, QNH, wind speed and direction, runway slope, fuel remaining or calculated take-off mass and aircraft configuration (+1).
3. **Application of Maximum Power:** Full engine power is to be applied with the throttle set to the take-off position. Engine stabilisation may be achieved either prior to or immediately following brake release, depending on aircraft performance characteristics and safety considerations.
4. **Simultaneous Brake Release and Timing Start:** The brakes shall be released, and the event marked simultaneously by the FTE via the event marker and lap trigger on the chronometer.
5. **Engine Parameter Recording:** The MAP and RPM values shall be recorded during the early acceleration phase to assist in later power analysis. This may involve a slight delay to ensure accurate manual reading if not logged digitally.
6. **Call-Out of Rotation Speed (V_R):** Upon reaching the predetermined rotation speed (approximately 46 KIAS), the flight test pilot (FTP) shall call the speed aloud, for example "46 !". The FTE will mark this point with a lap on the chronometer and the event marker. The corresponding speed will be annotated after, since there will be not enough time in this instant, and the FTE will have one hand on the FTI controller and the other on the chronometer.
7. **Log the airspeeds:** At this point the FTE will log the groundspeed on the flight test card.
8. **Acceleration to climb speed:** The pilot shall accelerate the plane up to $1.3 \cdot V_{S1}$, which corresponds to 63 KIAS in the SW121 and 56 KIAS for the SW128.
9. **Call-Out of 50-ft Altitude Speed (V_{50}):** When the aircraft reaches 50ft AGL, referencing either a visual ground cue or the altimeter, the pilot will call the airspeed "63 !". The FTE will record the lap time, the event marker and groundspeed.

10. **Engine data:** Finally, the FTE shall record the engine data once again in order to allow for the calculation of the average power during the takeoff manoeuvre. Record RPM and MAP or electric Power.

Data Gathering / Measurement

In the following table are shown the data measurements to be gathered during the TO testing from the cockpit of the plane. It is worth noting that all the data mentioned will be logged as well from both the flight computer and the additional FTI. The last of these will be most probably used ultimately for the data regression and analysis, but the live data gathering remains of crucial importance in case of a FT system malfunction or of a failed set-up.

The table shows the data gathered **prior to the test**, while on the runway.

QNH [ft]	QFE [ft]	OAT [K]	Wind speed [kn]	Wind head [deg]
		_____ + 273.15		

The table shows the data gathered during the test ranked based on importance of the data.

Test Point / Data	Full Power	Rotation	Liftoff	Obstacle (50ft)
Time [s]	0			
GS [kn]	0			
RPM [1/min]		/	/	
MAP (sw121) [inHg]		/	/	
Power (sw128) [kW]		/	/	

Takeoff Performance Calculation and Correction Methods

The methodology for calculating take-off distance and performing data regression is outlined in paragraph 4.4.2 of reference [2], elaborated further in paragraphs 23.53 and 23.59 of reference [4], and supplemented by the theoretical framework presented in the Kimberlin textbook [7]. These references provide guidance on how to correct the raw data acquired during take-off testing to account for the effects of wind and runway slope. Additionally, they define the various reference distances associated with the take-off manoeuvre. In accordance with the certification standards, the **total take-off distance** is defined as the horizontal distance required for the aircraft to reach an altitude of 50 feet above ground level, while maintaining a climb speed of not less than 1.3 times the stall speed in take-off configuration.

1. Total Take-Off Distance Calculation

The total take-off distance is determined under the assumption of constant acceleration between measurement points and is calculated as the sum of the individual segments of the take-off run and climb-out. The computational method used is presented below.

On-Ground Acceleration Calculation, assuming liftoff happens about 2 seconds after rotation:

Climb segment calculation:

$$a_{cl} = \frac{V_{50} - V_{LOF}}{t_{50} - t_{LOF}}, \quad S_2 = \frac{1}{2} a_{cl} \cdot (t_{50} - t_{LOF})^2$$

The total (uncorrected) take-off distance:

$$S_{TO} = S_g + S_{cl}$$

Remarks

Engine data — including MAP, RPM, and electrical power — are essential for reconstructing the thrust profile throughout the take-off sequence and should be recorded for each test point.

2. Wind Correction Formula

Following the formula to take into account the wind contribution on takeoff performance.

$$S_g = S_{g_{Test}} * (1 \pm V_w/V_{LOF})^{1.85}$$

Where:

S_g – no-wind take-off ground distance.

$S_{g_{Test}}$ – takeoff ground distance measured in testing.

V_w – wind velocity component parallel to the strip. + is used for headwind and – for tailwind.

V_{LOF} – groundspeed at lift-off with a known wind velocity.

3. Runway Slope Correction Formula

Following the formula to take into account the runway slope contribution on takeoff performance.

$$S_g = S_{g_{Test}} / [1 \pm (2 * g * S_{g_{Test}} / V_{LOF}^2) * \sin(\beta)]$$

Where:

$S_{g_{Test}}$ – ground distance measured during testing.

β – angle of the slope in radians. + for upslope and

g – acceleration of gravity.

– for downslope.

V_{LOF} – groundspeed at lift-off.

5. Shaft Power Calculation

SW121: the shaft power needs to be calculated from the Rotax 912 ULS manual through the values of RPM and MAP.

SW128: the shaft power needs to be calculated from the electric power exiting the batteries and engine RPM, knowing the efficiencies of the Electronic Speed Controller (ESC) (η_{esc}), the motor (η_m) and the cables (η_{cbl}). The following formula might be used: $P_b = P_{el} / (\eta_m * \eta_{cbl} * \eta_{esc})$. Said efficiencies can be found in the manufacturer's component manuals.

4. Weight and Power Correction Formulae

Following the formulae to take into account the variation in power, weight and air density compared to standard conditions.

$$\begin{cases} S_{1Std} = S_{1Test} \left(\frac{W_{Std}}{W_{Test}} \right)^{2.6} \left(\frac{\sigma_{Test}}{\sigma_{Std}} \right)^{1.9} \left(\frac{N_{Test}}{N_{Std}} \right)^{0.8} \left(\frac{P_{bTest}}{P_{bStd}} \right)^{0.6} \\ S_{1Std} = S_{1Test} \left(\frac{W_{Std}}{W_{Test}} \right)^{2.6} \left(\frac{\sigma_{Test}}{\sigma_{Std}} \right)^{1.9} \left(\frac{N_{Test}}{N_{Std}} \right)^{0.8} \left(\frac{P_{bTest}}{P_{bStd}} \right)^{0.6} \end{cases}$$

- Where:
- S_{1Test}, S_{2Test} – Calculated segment distances during the test condition. Ground roll and airborne distance respectively.
 - S_{1Std}, S_{2Std} – Corrected segment distances under standard conditions.
 - W_{Test} – Aircraft weight during the test.
 - W_{Std} – Aircraft weight under standard conditions.
 - σ_{Test} – Air density ratio during the test.
 - σ_{Std} – Air density ratio under standard conditions.
 - N_{Test} – Engine RPM for the takeoff manoeuvre during the test.
 - N_{Std} – Engine RPM for the takeoff manoeuvre under standard conditions.
 - P_{bTest} – Engine shaft power during the test.
 - P_{bStd} – Engine shaft power under standard conditions.

CHECKLIST – Test Procedure

To conclude, it follows a shared checklist / procedural step-by-step guide to be approved and discussed with the Flight Test Pilot. The aim of the checklist is to provide clear procedural guidelines for both FTE and FTP, in order to ensure precision and high efficiency in the highly dynamic test environment. This should mitigate the risk of a potential loss of data or failed acquisition.

FTP: Flight test pilot

FTE: Flight test engineer

Order	Action	Responsible	Notes
1	Line-up with runway	FTP	Flap position +1. Set altimeter to QFE .
2	Record runway conditions	FTE	QNH, QFE, OAT, wind speed, wind head, TO weight, config.
3	Full power	FTP	
4	Record engine data	FTE	RPM and MAP / Power.
5	Break release	FTP	Call aloud break release.
7	Time start	FTE	Event Mark + Start Chronometer.
8	Rotation	FTP	Call aloud the rotation speed (IAS).
9	Rotation data log	FTE	Chronometer lap + event marker.
10	Record speeds	FTE	Log groundspeed in test card (rotation).
11	Climb to 50ft	FTP	Call aloud the climbing speed at 50ft.
12	Record climb data	FTE	Chronometer lap + event marker.
13	Record 50ft speed	FTE	Log groundspeed at 50ft.
14	Record engine data	FTE	RPM and MAP / Power.
15	Record times	FTE	Log in the test card the three time laps from chronometer (rotation + liftoff + obstacle).

2.3 Longitudinal dynamic stability - Phugoid

General Objective

The objective of this test is to evaluate the aircraft's long-period dynamic stability, commonly referred to as the phugoid mode. This longitudinal oscillation is typically characterized by slow variations in pitch attitude, altitude, and airspeed, and it is generally lightly damped. In some cases, the motion may exhibit mild instability. According to CS-23, paragraph 23.181(c) [5], this level of instability is acceptable provided it does not significantly interfere with normal piloting tasks, such as trimming to a selected airspeed, maintaining altitude, or tracking a glide slope. The ASTM F2245, paragraph 4.5.6, specifies that the test must be repeated a sufficient number of times to ensure consistency and must be performed through several speeds spanning from $1.1 * V_{S1}$ up to the maximum speed (configuration specified in POH). Although, due to time limitation and aircraft autonomy constraints (SW128) the test will be conducted at cruising speed of 80 KIAS.

Remark: The airspeed will vary from the reference trim value by approximately $\pm 10\%$, or potentially slightly more. Therefore, it is essential to ensure that the aircraft remains within its operational and airspace limitations for the selected configuration (flaps 0). The expected airspeed range during the phugoid oscillation will be inside the margins [70 – 90] KIAS. Said speed range is considered safely flyable in the clean configuration (flaps 0) for both the SW121 and SW128, as indicated in their respective Pilot's Operating Handbooks [6][7]. Additionally, using the same test speed for both aircraft enables a direct comparison of their dynamic stability characteristics.

The evaluation will be performed under two configurations: **stick-fixed** and **stick-free**. To initiate the phugoid, the pilot will apply a gentle elevator input intended to create a speed deviation of approximately +10 KIAS from the trim condition. A slow pulse is preferred to avoid exciting higher-frequency modes. Once the desired speed variation is reached, the elevator control is returned to its initial position and either held in its equilibrium position (for stick-fixed) or released (for stick-free), allowing the aircraft to oscillate freely.

Each test point will be performed **twice**, once initiating the oscillation through an acceleration of +10 KIAS from the trim speed and the other through a deceleration of -10 KIAS from the trim speed. The **total amount of points are 4** and are divided as follows:

- Trim at 80 KIAS, Pulse to 90 KIAS (+10%), stick released to neutral position (free)
- Trim at 80 KIAS, Pulse to 70 KIAS (-10%), stick released to neutral position (free)
- Trim at 80 KIAS, Pulse to 90 KIAS (+10%), stick held (with control fixer) to neutral position (fixed)
- Trim at 80 KIAS, Pulse to 70 KIAS (-10%), stick held (with control fixer) to neutral position (fixed)

Throughout the oscillation, the pilot shall not intervene on the longitudinal control, unless safety limits (airspeed, load factor, etc.) are approached but is allowed to correct the airplane attitude along the lateral and directional axes. The FTE shall record the response in pitch attitude, airspeed and altitude using on-board instrumentation and the FTI system.

Certification Requirements / Tolerances

According to certification guidance:

- The phugoid mode is considered **neutrally stable** if the oscillation period is **less than 15 seconds**. In such cases, although the oscillation may not visibly damp out, it is still acceptable if it does not grow in amplitude or interfere with basic piloting tasks.
- To demonstrate **acceptable dynamic stability**, the **time to double the amplitude** of the oscillation must be **greater than 55 seconds**. This criterion ensures that any tendency toward instability develops so slowly that it does not adversely affect control or safety during normal flight operations.

- A **±10% deviation in airspeed** from the trimmed condition is considered **sufficient** to initiate a representative phugoid oscillation. The aircraft should be disturbed from its trimmed flight path with a slow, deliberate elevator input, and then allowed to oscillate freely without further control input.

Measurements and Data Acquisition

The phugoid motion evolves slowly enough to allow the recording of minimum and maximum airspeed excursions over time. This enables the construction of a response envelope from which the time to half or double amplitude can be determined [4].

The set of parameters to be recorded during the test includes IAS, PA, OAT, normal acceleration, AoA, pitch attitude, GPS speeds, engine power or MAP, and engine RPM.

Additional flight data will be acquired through the onboard instrumentation and flight test instrumentation systems. Where applicable, redundancy between manual recorded inputs and digital logging is implemented to improve data reliability and to allow cross-verification in the event of instrumentation malfunctions or inconsistencies.

The following tables summarise the minimum required data and corresponding sensor identifiers needed to ensure the accuracy and consistency of the phugoid stability tests performed on both the SW121 and SW128 aircraft.

Measurement	Type	Flight Card	Data - SW128	Data - SW128
Time	Required	UTC from FTI display and Event Marker	FTI GPS time	FTI GPS time
Airspeed	Required	Record from mechanical ASI	GSU 25C ADAHRS	Kanardia AHRS
PA	Required	Only if different from planed (from ALT)	GSU 25C ADAHRS	Kanardia AHRS
OAT	Required	Record from cockpit OAT sensor	GSU 25C ADAHRS	Kanardia AHRS
Ground Speed	Secondary	FTI GPS speed from FTI display	FTI GPS speed	FTI GPS speed
Fuel Flow	Secondary	Record from cockpit EIS	GEA 24 EIS	EPSI570C
Engine RPM	Secondary	RPM noted from engine display	GEA 24 EIS	EPSI570C
Engine MAP	Secondary	MAP noted from engine display	GEA 24 EIS	EPSI570C
AoA	Secondary	None	GSU 25C ADAHRS	Kanardia AHRS
Pitch Angle	Secondary	Artificial Horizon	GSU 25C ADAHRS	Kanardia AHRS

TABLE: PHUGOID MEASUREMENTS.

Test Method

The dynamic stability test points for the phugoid will be conducted in clean **configuration (flaps 0)** at a pressure altitude of **3’000 ft**. The test focuses on evaluating the aircraft’s dynamic stability at a single airspeed selected to be representative of the cruise condition for maximum range. This speed has been chosen to ensure that the aircraft **remains within its operational limits** in the specified configuration and that the test can be safely and consistently performed on both the SW121 and SW128. This approach allows for a direct comparison of the results during post-flight analysis.

All test points will be executed using both stick fixed and stick free techniques. To maximise overall test efficiency, the phugoid manoeuvres will be **conducted during transit segments**—either en route to the stall test location or on the return leg to the airfield. Since phugoid testing primarily requires straight-line flight

over several kilometres, performing these points during transfer flights is an effective way to **optimise mission time, reduce fuel consumption, and lower operational costs.**

The following outlines the detailed procedure for executing each test point:

1. **Trim shot:** the pilot shall climb to 3'000 ft, set the aircraft flaps to 0 position and perform a trim shot at 80 KIAS.
2. **Condition acquisition:** the FTE shall record the trim shot data: IAS, PA, OAT, RPM, MAP or Power, event marker number, configuration and weight (SW121).
3. **Event marker:** the FTE shall now flag the moment with an event marker + START the chronometer.
4. **+10 KIAS - free:** the pilot shall pitch down the aircraft slowly (min 10 sec) without touching any other input until the airspeed reaches 90 KIAS. Once the airspeed has reached said value, the pilot shall release completely the stick leaving that it returns to the neutral position. The pilot is **allowed to correct the aircraft attitude in the lateral and directions.**
5. **Data acquisition:** the FTE shall record in the flight test card the dynamics of the oscillation, primarily recording **airspeed** and **time**. In detail, the FTE shall record each peak or valley of the oscillation with the event marker and additionally writing down the airspeed and chronometer LAP time. Repeat until the oscillation is no longer perceptible.
6. **Trim shot:** the pilot shall climb to 3'000 ft, set the aircraft flaps to 0 position and perform a trim shot at 80 KIAS.
7. **Condition acquisition:** the FTE shall record the trim shot data: IAS, PA, OAT, RPM, MAP or Power, event marker number, configuration and weight (SW121).
8. **Event marker:** the FTE shall now flag the moment with an event marker + START chronometer.
9. **-10 KIAS - free:** the pilot shall pitch up the aircraft slowly (min 10 sec) without touching any other input until the airspeed reaches 70 KIAS. Once the airspeed has reached said value, the pilot shall release completely the stick leaving that it returns to the neutral position. The pilot is now only allowed to correct the aircraft attitude in the lateral and directions (unless to maintain flight safety).
10. **Data acquisition:** the FTE shall record in the flight test card the dynamics of the oscillation, primarily recording **airspeed** and **time**. In detail, the FTE shall record each peak or valley of the oscillation with the event marker and additionally writing down the airspeed and chronometer LAP time. Repeat until the oscillation is no longer perceptible.
11. **Condition acquisition:** the FTE shall record once again the following data: PA, OAT, RPM, MAP or Power, event marker number and weight (SW121).

STICK FIXED PHUGOID TESTING

The stick-fixed phugoid test is conducted in the **same manner as the stick-free test described above**, with one key difference: after increasing or decreasing the airspeed by approximately 10 KIAS **the pilot shall return with the stick to the equilibrium position and HOLD it in place throughout the oscillation.** To accurately identify this equilibrium point, the **control fixer** instrument is employed. This device consists of a rubber band placed on the FTP leg. Once the aircraft is properly trimmed, the rubber band is positioned to mark the neutral stick location. After initiating the test by applying the appropriate input ($\pm 10\%$ of the trimmed airspeed), the pilot must return the control stick to the marked position as indicated by the **control fixer**.

Data Gathering / Measurement

In the following tables are shown the data measurements to be gathered onto the *flight test card*. It is worth noting that all the data mentioned will be logged as well from both the flight computer and the additional FTI. The main quantities used in post-processing for data analysis will be the recorded time and airspeed from

the flight computer. Although the flight card test data is used as hard backup in case of failure of the system of in setting the system up.

Table to be filled just **after trim shot**.

QNH [ft]	OAT [K]	Weight [kg] (SW121)	Configuration	Event number (last)
	_____ + 273.15			

Table to be filled with data from each one **phugoid test point**.

Test Point	1 st Peak	2 nd Peak	3 rd Peak	4 th Peak	5 th Peak	6 th Peak
Time UTC						
Airspeed [KIAS]						

Phugoid Performance Calculation and Correction Methods

Phugoid testing is conducted under the assumption of steady wind conditions, with constant speed and direction. While atmospheric wind is naturally present during flight, it is not directly factored into the analysis, as it does not significantly influence the intrinsic dynamic stability characteristics of the aircraft. To evaluate the phugoid mode, the aircraft’s response to an initial longitudinal disturbance is analysed. The primary objective is to determine whether the resulting oscillatory motion is stable—where the amplitude diminishes over time—or lightly unstable, in which case the amplitude grows progressively. In both scenarios, the time-domain characteristics of the oscillation must be assessed, specifically the oscillation **period** and, if applicable, the **time to double** amplitude.

According to certification guidance [5], an oscillation period of approximately **15 seconds** is indicative of neutral stability. In the case of lightly unstable behaviour, a time to double amplitude greater than **55 seconds** is deemed acceptable, reflecting adequate longitudinal dynamic stability for normal piloting tasks.

To quantify these properties, airspeed (CAS) and AoA should be plotted as a function of time, and the characteristic oscillation period determined from the waveform. A representative example of this analysis procedure is shown in Figure 5.

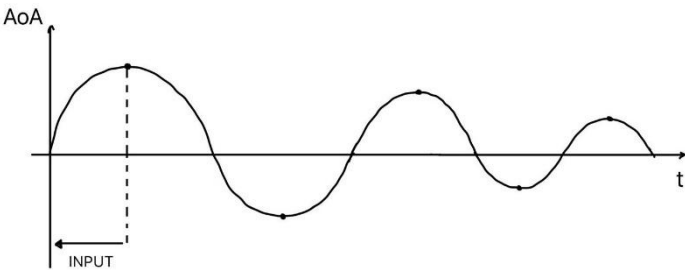


FIGURE 5: PHUGOID DATA ANALYSIS.

The only correction method that should be applied to the data is the correction from IAS, gathered from the ASI, to corresponding CAS value, following the correction curve described in the POHs [5][6].

CHECKLIST – Test Procedure

The following checklist outlines a shared procedural guide to be reviewed and approved by the Flight Test Pilot. It is intended to ensure clarity, precision, and operational efficiency for both the FTE and FTP during testing. The structure aims to minimise the risk of data loss or failed acquisition. The checklists are divided into **stick free phugoid testing** and **stick fixed phugoid testing**.

FTP: Flight test pilot

FTE: Flight test engineer

Table for the **stick free** phugoid testing.

Order	Action	Responsible	Notes
1	Reach 3'000 ft PA	FTP	Configuration clean (0).
2	Trim shot	FTP	Trim shot at 80 KIAS, 3'000 ft.
3	Record conditions	FTE	OAT, PA, RPM, MAP or Power, last event nr., config. and weight (SW121).
4	Event marking	FTE	Mark the event with the FT controller + Chronometer START.
5	90 KIAS - Free	FTP	Dive to 90 KIAS in a pulse of 10 seconds and LEAVE the stick to go back to neutral position.
6	Aircraft Control	FTP	Lateral and directional control is allowed.
7	Record acquisition	FTE	Mark peaks and valleys with event marker, chrono LAP and IAS.
8	Trim shot	FTP	Trim shot 80 KIAS, 3'000 ft.
9	Record conditions	FTE	OAT, PA, RPM, MAP or Power, last event nr., config. and weight (SW121).
10	Event marking	FTE	Mark the event with the FT controller + Chronometer START.
11	70 KIAS - Free	FTP	Pull to 70 KIAS in a pulse of 10 seconds and LEAVE the stick to go back to neutral position.
12	Aircraft Control	FTP	Lateral and directional control is allowed.
13	Record acquisition	FTE	Mark peaks and valleys with event marker, chrono LAP and IAS.
14	Record conditions	FTE	OAT, PA, RPM, MAP or Power, last event nr., config. and weight (SW121).

Table for the **stick fixed** phugoid testing.

Order	Action	Responsible	Notes
1	Reach 3'000 ft PA	FTP	Configuration clean (0).
2	Trim shot	FTP	Trim shot at 80 KIAS, 3'000 ft.
3	Control fixer	FTP	Set up the control fixer to the neutral position
4	Record conditions	FTE	OAT, PA, RPM, MAP or Power, last event nr., config. and weight (SW121).
5	Event marking	FTE	Mark the event with the FT controller + Chronometer START.
6	90 KIAS - Free	FTP	Dive to 90 KIAS in a pulse of 10 seconds and BRING the stick to go back to neutral position and HOLD IN POSITION using control fixer.
7	Aircraft Control	FTP	Lateral and directional control is allowed.

8	Record acquisition	FTE	Mark peaks and valleys with event marker, chrono LAP and IAS.
9	Trim shot	FTP	Trim shot at 80 KIAS, 3'000 ft.
10	Record conditions	FTE	OAT, PA, RPM, MAP or Power, last event nr., config. and weight (SW121).
11	Event marking	FTE	Marke the event with the FT controller + Chronometer START.
12	70 KIAS - Free	FTP	Pull to 70 KIAS in a pulse of 10 seconds and BRING the stick to go back to neutral position and HOLD IN POSITION using control fixer .
13	Aircraft Control	FTP	Lateral and directional control is allowed.
14	Record acquisition	FTE	Mark peaks and valleys with event marker, chrono LAP and IAS.
15	Record conditions	FTE	OAT, PA, RPM, MAP or Power, last event nr., config. and weight (SW121).

2.4 Stall Performance - Flap 0

General Objective

The objective of the stall testing is to evaluate the stall behaviour and performance of both aircraft, the SW121 and SW128, in **clean configuration (flaps retracted)**, and to verify compliance with applicable certification requirements. This assessment will also provide comparative insights into the aerodynamic characteristics of the two aircraft.

The test methodology is outlined in *paragraph 4.4.1 of ASTM F2245 – 12d [2]*, which specifies that stall speed V_S must be determined by flight testing under the following conditions: **rate of speed decrease not exceeding 1 kt/s, throttle closed, maximum take-off weight, and most unfavourable centre of gravity** (corresponding to the most aft position). **Although** it is important to notice that this CG condition is in conflict with the one requested for the takeoff testing. For this reason, as previously mentioned in paragraph 2.1, the CG position for testing will be set to an intermediate value: **SW121 CG: 30.5 %. SW128 CG: 30 %**. Additionally, the criteria of *CS-23 Amendment 4, paragraphs 23.49 and 23.201 [4]*, will be observed.

Stall points will be flown at **idle power**, with **wings level**, and a minimum of **three deceleration rates** will be used: -0.5 kt/s, -2 kt/s, and -3 kt/s. These values are chosen to allow interpolation around the target certification condition of -1 kt/s. To minimise the effect of wind, **each test point will be performed in reciprocal headings** (twice).

The stall will be considered achieved and recorded when any of the following occurs:

- An uncontrollable downward pitching motion.
- A pitch-down motion triggered by an automatic system (e.g. stick pusher).
- Control deflection reaching its mechanical stop.

Summarizing, a total, **six primary stall points** are planned (each deceleration planned twice at reciprocal headings). To ensure test reliability and allow for the repetition of any points with unsatisfactory results, up to **four extra attempts** may be conducted, bringing the total number of planned test points to **ten**.

Given the nature of stall manoeuvres, a **loss of altitude** during each entry is expected. However, based on the Pilot Operating Handbooks (POHs) for the SW121 and SW128 [5][6], this loss is not anticipated to exceed **250 ft**, even under worst-case conditions (maximum weight and slow recovery). Such altitude loss can be recovered between test points, but altitude margins must be monitored continuously to ensure **operational safety** and compliance with **zone airspace restrictions** throughout the mission.

Before initiating the test sequence, the **wind direction and intensity at the testing altitude** will be determined using the “*wind cloverleaf*” method. This procedure is essential for accurate post-test data reduction and for identifying wind influence on the results. Ideally, the wind cloverleaf should be conducted both **before and after** the test series to assess any change in wind conditions during the flight. However, due to the **limited flight endurance**, particularly of the electric-powered SW128, the second wind cloverleaf may be omitted. In this case, it must be assumed that wind conditions remained reasonably constant throughout the test window.

Certification Requirements / Tolerances

In accordance with the certification requirements outlined in references [2] and [4], the following criteria must be met during stall testing to demonstrate regulatory compliance:

- The aircraft shall be considered stalled only in one of the following behaviours: uncontrollable downward pitching motion, downward pitch triggered by an automated stall-protection device, control input reaches the mechanical stop without further effect.
- During **each stall** entry and recovery, the *test pilot* must confirm the following:
 - The **stick force gradient remains positive** up to the stall (i.e. a pulling force is required).
 - The aircraft remains controllable in **roll and yaw** up to the stall using **unreversed** control inputs.
- The **rate of speed reduction** during stall entry must not exceed **1 knot per second**, as prescribed in paragraph 23.201(b) of CS-23.
- The extent of **roll and/or yaw** encountered during stall recovery is documented.

Measurements and Data Acquisition

The minimum set of parameters to be acquired during the test includes: IAS, PA, OAT, aircraft attitude angles (pitch, roll, yaw), GPS-derived speeds, in particular the ground speed (V_{GS}), deceleration rate, engine power or MAP, engine RPM, elevator position, and longitudinal control force (based on pilot feedback).

Additional flight data will be recorded via the on-board instrumentation and FTI systems. Where appropriate, redundancy between manual input and digital logging is implemented to enhance data reliability and allow cross-verification in the event of instrumentation failure or anomalies.

The following tables summarise the minimum required data, along with corresponding sensor codes, necessary to ensure the validity and consistency of the stall tests conducted on both the **SW121** and **SW128** aircraft.

Measurement	Type	Flight Card	Data - SW121	Data - SW128
Time	Required	UTC from FTI display and Event Marker	FTI GPS time	FTI GPS time
Groundspeed	Required	Record from left-hand PFD	GDU 470	Kanardia AHRS
PA	Required	Only if different from planed (ALT)	GSU 25C ADAHRS	Kanardia AHRS
OAT	Required	Record from cockpit OAT sensor	GSU 25C ADAHRS	Aera 660
Ground Speed	Required	FTI GPS speed from FTI display	FTI GPS speed	FTI GPS speed
Entry Rate	Required	FTI display (set on entry rate screen)	FTI GPS speed	FTI GPS speed
Fuel Flow	Required	Record from cockpit EIS	GEA 24 EIS	EPSI570C
Engine RPM	Required	RPM noted from engine display	GEA 24 EIS	EPSI570C
Engine MAP	Required	MAP noted from engine display	GEA 24 EIS	EPSI570C

Stick Force	Required	From pilot feedback	None	None
Elevator Position	Secondary	None	Potentiometer	Potentiometer
AoA	Secondary	None	GSU 25C ADAHRS	Kanardia AHRS
AoS	Secondary	None	GSU 25C ADAHRS	Kanardia AHRS

TABLE 12: STALL MEASUREMENTS.

Test Method

The test will be carried out in the **flap-retracted (0) configuration**. The corresponding IAS - CAS correction method for post-processing will match this configuration. Prior to stall testing, the aircraft shall reach a designated altitude, preliminarily set at **3'000 ft**, subject to final confirmation by the pilot. Once level at this altitude, the *wind cloverleaf procedure* will be executed to determine wind speed and direction, enabling appropriate corrections to be applied during post-flight analysis.

Note: It is essential to remain close to the selected test altitude throughout all stall test points, to avoid significant variations in wind and temperature.

Wind Determination

The wind will be determined by flying a **square pattern** consisting of four straight legs, each separated by approximately 90°. The method assumes constant wind conditions over time and altitude throughout the test sequence. The detailed method to perform this manoeuvre is described as follows:

1. **Altitude reach:** Once reached the prescribed altitude, the FTE shall record: OAT, PA, weight, aircraft configuration (0), event marker number.
2. **Trim shot 1st leg:** the pilot shall maintain as constant as possible the IAS of the aircraft, and secondly also keeping constant the heading of flight for at least **1 minute**. **Example:** 80KIAS, 3'000 ft, 0 deg.
3. **1st Acquisition:** Once the condition is established, the pilot shall call the stabilized condition and try holding the condition. The FTE should record the event with the event marker, shall record the groundspeed (kt) and heading of the leg.
4. **Trim shot 2nd leg:** the pilot shall turn of about 90 degrees, maintaining altitude and perform the second trim shot, trying to aim primarily at the same airspeed as before, and secondly at keeping the heading constant for at least **1 minute**. **Example:** 80KIAS, 3'000 ft, 270 deg.
5. **2nd Acquisition:** Once the condition is established, the pilot shall call the stabilized condition and try holding the condition. The FTE should record the event with the event marker, shall record the groundspeed (kt) and heading of the leg.
6. **Trim shot 3rd leg:** the pilot shall turn again of about 90 degrees, maintain altitude and perform the third trim shot, trying to aim primarily at the same airspeed as before, and secondly at keeping the heading constant for at least **1 minute**. **Example:** 80KIAS, 3'000 ft, 180 deg.
7. **3rd Acquisition:** Once the condition is established, the pilot shall call the stabilized condition and try holding the condition. The FTE should record the event with the event marker, shall record the groundspeed (kt) and heading of the leg.
8. **Trim shot 4th leg:** the pilot shall turn again of about 90 degrees, maintaining altitude and perform the last trim shot, trying to aim primarily at the same airspeed as before, and secondly at keeping the heading constant for at least **1 minute**. **Example:** 80KIAS, 3'000 ft, 90 deg.
9. **4th Acquisition:** Once the condition is established, the pilot shall call the stabilized condition and try holding the condition. The FTE should record the event with the event marker, shall record the groundspeed (kt) and heading of the leg.

10. **Record final data:** finished the wind determination, the FTE shall record once again: OAT, PA, weight, aircraft configuration (0).

Note 1: Errors in trimmed IAS have a greater impact than variations in heading or altitude. Prioritise accurate speed trimming.

Note 2: Slight altitude variation between legs is acceptable, provided it remains within ± 250 ft.

Stall Speed Test

Once wind data has been acquired, the stall testing begins. The aircraft will fly a similar **square pattern**, ensuring that each entry rate test point is repeated in opposite directions. This ensures wind effects are minimised and allows for result comparison. The selected pattern supports:

- Operation within a known airspace envelope where wind and weather conditions are understood and stable.
- Efficient use of fuel/energy, enabling repetitions with minimal repositioning.
- Continuous wind estimation through analysis of trim conditions (less accurate than wind cloverleaf).
- Simple test point reordering in case of data loss or failed test points.
- Minimal change in CG between reciprocal test points.

Here a practical approach to the test technique and operational manoeuvres to perform:

1. **Reach altitude:** the pilot shall reach the prescribed altitude and start loitering. Meanwhile, the FTE shall record all the test conditions: PA, OAT, aircraft configuration (0), weight and event number.
2. **1st trim shot:** the pilot shall perform a trim shot at constant prescribed heading and airspeed equal to $V_{trim} = 1.5 * V_s$. Said airspeed corresponds to 76 KIAS for both the SW121 and the SW128.
3. **1st entry rate:** the pilot shall put the power setting to **idle** and try at aiming at a deceleration rate of 3 kn/s. The pilot shall fine-tune the “exchange” of the aircraft kinetic and geopotential energy in order to maintain said deceleration. The FTE shall help the pilot reach said deceleration target with the FTI data from the FTI display. The FTE shall record the entry rate and the engine RPM and engine power / MAP.
4. **1st stall call:** The pilot shall call aloud the airspeed the stall condition whenever one of the following happens: *uncontrollable downward pitching motion, downward pitch triggered by an automated stall-protection device, control input reaches the mechanical stop without further effect*. **For example: “50 !”**. At the same time the FTE shall record the event with the event marker and write down on the flight card the groundspeed and UTC time.
5. **Recovery:** the pilot shall now recover from the stall and reposition the aircraft at the prescribed altitude and heading for the following test point.

The procedure (ALL points above) shall be repeated for each of the stall test points (all entry rates).

Note: The pilot is only allowed to correct asymmetries using small roll and yaw correction inputs.

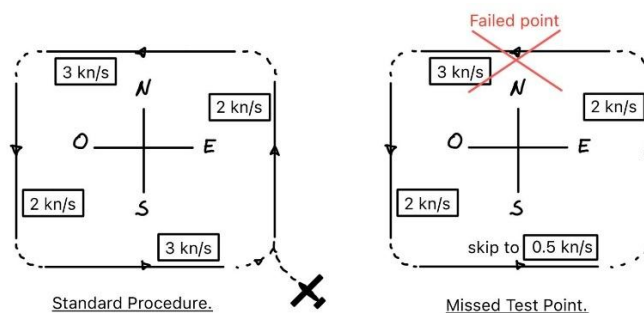


FIGURE 6: STALL TEST POINTS.

Data Gathering / Measurement

In the following tables are shown the data measurements to be gathered from the cockpit onto the *flight test card*. It is worth noting that all the data mentioned will be logged as well from both the flight computer and the additional FTI. The GPS speed coupled with the wind speed determination will be primarily used to calculate the TAS of the various test points. The reason behind this procedure is that at high values of AoA (stall) the aerodynamic field around the plane is usually highly distorted and unstable, making the ASI measurement less reliable. Although, the ASI measurement still remains in the flight card as a hard-backup measurements and can be a useful information for data post-processing and analysis.

The table shows the data gathered **once at testing location**:

QNH [ft]	OAT [K]	Weight [kg] (SW121)	Configuration	Event number (last)
	_____ + 273.15			

The table shows the data gathered during the **wind determination** (cloverleaf). **Note:** try to aim at 80 KIAS and keep it as constant as possible in between legs.

Test Point / Data	1 st leg	2 nd leg	3 rd leg	4 th leg
Airspeed [KIAS]				
Heading [deg]				

The table shows the data gathered **before stall testing**:

QNH [ft]	OAT [K]	Weight [kg] (SW121)	Configuration	Event number (last)
	_____ + 273.15			

The table shows the data gathered during the **stall speed testing**:

Test Point	HDG [deg]	V_{GS} [kn]	RPM [1/min]	MAP [inHg]
0.5				
0.5 - opposite				
2				
2 – opposite				
3				
3 - opposite				

Stall Performance Calculation and Correction Methods

Accurate evaluation of stall performance necessitates the application of appropriate correction methods to minimise data variability and standardise results across varying test conditions. This section outlines the procedures used to account for wind influence, weight deviations, and airspeed calibration errors. Among these, wind determination and correction represent the most critical adjustments required during stall testing. Given the significant aerodynamic disturbances present at low speeds and high angles of attack,

airspeed indicator (ASI) readings can be unreliable. Therefore, GPS-derived ground speed, corrected using the calculated wind speed and direction, is employed to determine the true stalling speed with greater accuracy.

Wind Calculation and Correction

Wind correction is derived from the data gathered during a dedicated wind determination manoeuvre, which assumes that both wind direction and magnitude remain constant throughout the four legs of the square pattern. The method applies the Carnot theorem to the geometry of the flight path, combined with the vectorial relationship: $V_{TAS} = V_w + V_{GS}$.

By applying this principle to each leg of the pattern, a solvable system of three equations with three unknowns—true airspeed (TAS), wind speed, and wind direction—can be constructed. The fourth leg offers redundancy, allowing the system to be solved three times using different combinations of three legs. The final wind vector is obtained by averaging these three solutions, which enhances robustness and reduces sensitivity to individual measurement errors. During post-processing, the previously determined wind vector is used to correct GPS-derived ground speed data from the stall manoeuvres.

Weight Correction

To enable consistent comparison of stalling speeds across different aircraft loading conditions, test results are corrected to a standard weight using the following relationship:

$$V_s = V_{s_{Test}} * (W_{std}/W_{Test})^{1/2}$$

Where:

V_s – weight-corrected airspeed (CAS)

W_s – standard weight

$V_{s_{Test}}$ – test stalling airspeed (CAS)

W_{Test} – testing weight

Result Extrapolation

To account for dynamic effects due to different approach rates to the stall, the corrected stall speeds are plotted against the measured rate of speed reduction (entry rate). A regression is then performed to interpolate the stall speed at a reference entry rate of 1 knot per second. This process ensures consistency in the evaluation of stall performance. A visual representation of this regression method is provided in Figure 7.

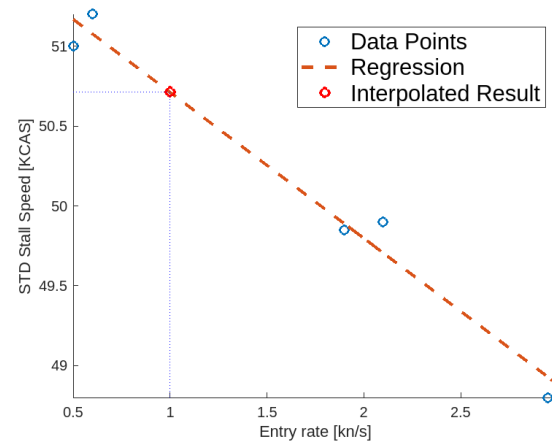


FIGURE 7: STALL DATA EXAMPLE.

Airspeed Correction

Finally, airspeed values recorded from the aircraft's airspeed indicator (IAS) are corrected to calibrated airspeed (CAS) using conversion tables provided in the aircraft's Pilot Operating Handbooks (POHs) [6][7], specifically those corresponding to the clean configuration. These corrections ensure fidelity between indicated and actual aerodynamic performance.

CHECKLIST – Test Procedure

The following checklist outlines a shared procedural guide to be reviewed and approved by the Flight Test Pilot. It is intended to ensure clarity, precision, and operational efficiency for both the FTE and FTP during

testing. The structure aims to minimise the risk of data loss or failed acquisition. The checklists are divided into **wind determination checklist** and **stall test checklist**.

FTP: Flight test pilot

FTE: Flight test engineer

For the **wind determination**.

Order	Action	Responsible	Notes
1	Reach 3'000 ft PA	FTP	Configuration clean (0).
2	Record conditions	FTE	OAT, PA, configuration, last event number and weight (SW121).
3	Trim shot – 1st leg	FTP	Constant IAS and heading and altitude (80 KIAS, 0°, 3'000 ft). 1 minute.
4	Record acquisition	FTE	Mark event, GS, heading.
5	Trim shot – 2nd leg	FTP	Turn ~90° (80 KIAS, 270°, 3'000 ft). 1 minute.
6	Record acquisition	FTE	Mark event, GS, heading.
7	Trim shot – 3rd leg	FTP	Turn ~90° (80 KIAS, 180°, 3'000 ft). 1 minute.
8	Record acquisition	FTE	Mark event, GS, heading.
9	Trim shot – 4th leg	FTP	Turn ~90° (80 KIAS, 90°, 3'000 ft). 1 minute.
10	Record acquisition	FTE	Mark event, GS, heading.
11	Record conditions	FTE	OAT, PA, configuration, last event number and weight (SW121).

For the **stall testing**.

Order	Action	Responsible	Notes
1	Reach 3'000 ft PA	FTP	Maintain altitude; configuration clean (0).
2	Record conditions	FTE	PA, OAT, aircraft configuration, last event number and weight (SW121).
3	Trim shot	FTP	Trim at constant heading and airspeed 76 KIAS.
4	Begin stall entry	FTP	Throttle idle; Target X kn/s deceleration using only pitch control.
5	Monitor deceleration	FTE	Support with real-time FTI data; Record entry rate, RPM, MAP / power.
6	Stall recognition call	FTP	Call airspeed aloud at first indication of stall (break, pusher or stop).
7	Record acquisition	FTE	Mark event; Record GS.
8	Recover	FTP	Return to test altitude.
9	Reposition	FTP	Turn 90° heading for next test point.

Note: This procedure is repeated for each stall entry rate point.

2.5 Landing performance

General objective

The landing test is conducted to evaluate the aircraft's minimum landing distance in accordance with applicable certification requirements. The primary regulatory references are CS-23, paragraph CS 23.75 [5], and ASTM, paragraph 4.4.4 [4].

The objective of the test is to assess the aircraft's ability to complete a stabilised approach and landing within the required distance limits, across varying configurations and environmental conditions, while maintaining effective control throughout the manoeuvre. The results will serve to validate the landing distance declared in the Pilot's Operating Handbook and to confirm compliance with certification standards.

It should be noted that there is a difference in landing configuration between the SW121 and the SW128, as outlined in their respective POHs. The SW121 uses a configuration with fully deployed flaps (+2) and partially extended airbrakes ($\frac{1}{2}$), whereas the SW128 employs only fully deployed flaps (+2), without airbrake use.

Certification Requirements / Tolerances

The certification frameworks [4][5] mention and pose some constraints on the landing testing technique and allowable results.

- The approach shall be stabilized before arriving to the 50 ft obstacle, this includes gliding angle, sink rate, airspeed and engine.
- The airspeed at the 50 ft obstacle (V_{50}) shall be at least $1.3 * V_{50}$. Operatively, this prescribes approach speeds of the SW121 and SW128 of at least **60 KIAS**.
- The SW121 throttle position shall be set to rich mixture.
- Both the flare and braking sections shall be performed smoothly and with the aircraft aligned with the runway.

Measurements and Data Acquisition

Due to the highly dynamic nature of the landing manoeuvre, the number of measured parameters has been intentionally limited to reduce operational complexity and minimise pilot and crew workload. To support accurate post-flight analysis and enable environmental corrections, a series of measurements must be collected and logged prior and after the test. These include OAT, DA, runway slope or inclination, wind speed, wind direction, and either the landing weight or the remaining fuel on board (SW121).

All remaining flight parameters will be acquired via the on-board instrumentation and flight test instrumentation systems. Where appropriate, redundancy is ensured by maintaining both manual and digital logging. This approach allows for cross-verification and enhances data integrity in the event of instrumentation faults or inconsistencies.

Additionally, to further increase redundancy, GPS data from the aircraft's native systems will be recorded throughout the test. For this purpose, data will be acquired from the GNC 355 in the SW121 and from the Aera 660 in the SW128.

Measurement	Type	Flight Card	Data - SW121	Data - SW128
Time	Required	UTC on FT display and Event Marker	FTI GPS time	FTI GPS time
Speed	Required	GPS speed on FT screen	FTI GPS speed	FTI GPS speed
Distance	Required	Visual marks on the runway	FTI GPS position	FTI GPS position
Glide Slope	Required	None	FTI GPS speed	FTI GPS speed
Engine RPM	Secondary	RPM noted from engine display	GEA 24 EIS	EPSI570C
Engine MAP	Secondary	MAP noted from engine display	GEA 24 EIS	EPSI570C

Height	Secondary	From ALT, both QNH and QFE	GSU 25C + GPS FTI	AHRS + GPS FTI
Elevator Pos.	Secondary	None	Potentiometer	Potentiometer

TABLE 13: LND MEASUREMENTS.

Test Method

The pilot shall make sure to be descending towards the runway and in the configuration described in the POH of both planes. The landing configuration for the SW121 corresponds to full flaps (+2) and half airbrakes (1/2) deployed. For the SW128, the landing configuration only prescribes full flap deployment (+2).

- Altimeter regulation:** Set altimeter to **QFE regulation** for correctly visualizing the 50 ft condition.
- Stable descent:** the pilot shall aim at stabilizing the descent towards the runway at a speed of at least 60 KIAS.
- Condition log:** the FTE shall record all the relevant environmental and condition variables, such as plane configuration, event marker number, approach groundspeed, RPM and MAP or Power and weight (SW121).
- Obstacle:** the pilot shall reach the obstacle holding the prescribed 60 KIAS speed and call aloud the airspeed (V_{50}).
- Data acquisition:** the FTE shall be focused on the plane’s surroundings and visual reference points. The FTE shall flag the event on the event marker, as well as start the chronometer. Consequently, write down the approach groundspeed and the visual reference point.
- Touchdown:** the pilot shall touch down on the runway maintaining an aligned heading. The pilot shall call aloud the touchdown airspeed and start breaking.
- Data acquisition:** the FTE shall flag the event with the event marker and lap the chronometer. Record groundspeed and visual reference point.
- Full stop:** the pilot shall bring the plane to a full stop.
- Data acquisition:** the FTE shall record the stop event with an event marker and lap the chronometer. Record the visual distance marker
- Condition log:** finally, once completely landed the FTE shall record the condition just after the landing. This includes OAT, QNH, QFE, weight (SW121) and the wind conditions on the runway (direction and intensity).

Data Gathering / Measurement

In the following table are shown the data measurements to be gathered during the landing testing from the flight test card. It is worth noting that all the data mentioned will be logged as well from both the flight computer and the additional FTI. The last of these will be most probably used ultimately for the data regression and analysis, but the live data gathering remains of crucial importance in case of a FT system malfunction or of a failed set-up.

Data to be gathered **prior to the test**, while performing the long approach.

Configuration	RPM [1/min]	MAP [inHg]	Power [kW]	LND weight [kg]	Event marker nr.

The table shows the data gathered during the landing

Test Point / Data	Descent	Obstacle (50 ft)	Touchdown	Full stop
Groundspeed [kt]				0

Time [s]	/	0		
Distance [vis. Ref.]				
RPM [1/min]		/	/	
MAP (SW121) [inHg]		/	/	
Power (SW128) [kW]		/	/	

Data to be gathered **consequently to the test**, having just landed.

QNH [ft]	QFE [ft]	OAT [K]	Wind speed [kn]	Wind head [deg]	LND weight [kg]
		_____ + 273.15			

Landing Performance Calculation and Correction Methods

The final data used for the landing distance assessment will be the GPS data acquired from the additional FTI system. However, this data is influenced by several factors and must be corrected and standardised to ensure comparability with other test results. Specifically, the raw data must be adjusted to account for wind conditions, runway slope, aircraft weight, and air density. The procedures described and followed in this section are known as the USN and USAF TPS methods [9].

Following a simplified procedure to reduce the data gathered from the flight test cards to obtain an approximate, but sufficient precision data on the landing distance. The total landing distance is determined under the assumption of constant acceleration between measurement points and is calculated as the sum of the individual segments of the take-off run and climb-out. The computational method used is presented below. All the speeds presented are groundspeeds from GPS.

Descent segment calculation, considering only the horizontal component of the deceleration:

$$a_{ds} = \frac{V_{50} - V_{TD}}{t_{50} - t_{TD}}, \quad S_3 = \frac{1}{2} a_{ds} \cdot (t_{50} - t_{TD})^2$$

On-Ground Acceleration Calculation

$$a_g = \frac{V_{TD}}{t_{TD} - t_{stop}}, \quad S_g = \frac{1}{2} a_g \cdot (t_{TD} - t_{stop})^2 = S_4$$

The total (uncorrected) take-off distance:

$$S_{TO} = S_3 + S_4$$

Slope correction

The runway slope correction is identical to the takeoff corresponding correction and described in reference [4].

$$S_{4Std} = S_{4Test} / [1 \pm (2gS_{4Test}/V_{TD}^2) * \sin(\beta)]$$

Where:

S_{4Test} – ground distance measured during testing.

g – acceleration of gravity.

V_{TD} – airspeed at touchdown.

β – angle of the slope in radians. + for downslope and – for upslope.

Wind correction

The wind correction is performed applying the following formulae from reference [9].

$$\begin{cases} V_{TD} = V_{TD_{Test}} - V_w \\ S_{4_{Std}} = S_{4_{Test}} \left(1 + \frac{V_w}{V_{TD}} \right)^{1.85} \end{cases}$$

Where the variables flagged with the X_{Test} are the ones measured / obtained from post-processing of the FTI GPS data log.

Weight and density correction

The difference in weight and air density with respect to the standard conditions can be corrected following the procedure proposed by reference [9].

$$\begin{cases} E_h = \frac{V_{50}^2 - V_{TD}^2}{2g} \\ S_{3_{Std}} = S_{3_{Test}} \left(\frac{W_{Std}}{W_{Test}} \right)^{\left(2 + \frac{E_h}{E_h + 50} \right)} * \left(\frac{\sigma_{Test}}{\sigma_{Std}} \right)^{\left(\frac{E_h}{E_h + 50} \right)} \\ S_{4_{Std}} = S_{4_{Test}} \left(\frac{W_{Std}}{W_{Test}} \right)^2 \left(\frac{\sigma_{Test}}{\sigma_{Std}} \right) \end{cases}$$

Where E_h is the Energy Height, σ is the air density ratio for a given condition and W_{Test} , W_{Std} are the landing weight and standard weight respectively. Finally, the complete landing distance can be found by summing the two contributions (airborne and ground).

$$S_{LND} = S_{3_{Std}} + S_{4_{Std}}$$

Shaft Power Calculation

SW121: the shaft power needs to be calculated from the Rotax 912 ULS manual through the values of RPM and MAP.

SW128: the shaft power needs to be calculated from the electric power exiting the batteries and engine RPM, knowing the efficiencies of the Electronic Speed Controller (ESC) (η_{esc}), the motor (η_m) and the cables (η_{cbl}). The following formula might be used: $P_b = P_{el} / (\eta_m * \eta_{cbl} * \eta_{esc})$. Said efficiencies can be found in the manufacturer's component manuals.

CHECKLIST – Test Procedure

The following checklist outlines a shared procedural guide to be reviewed and approved by the Flight Test Pilot. It is intended to ensure clarity, precision, and operational efficiency for both the FTE and FTP during testing. The structure aims to minimise the risk of data loss or failed acquisition.

FTP: Flight test pilot

FTE: Flight test engineer

Order	Action	Responsible	Notes
1	Landing Config.	FTP	SW121: flaps +2, airbrakes ½; SW128: flaps +2 only. Set altimeter to QFE .
2	Stabilised Descent	FTP	Maintain approach speed ≥ 60 KIAS, long approach.

3	Log conditions	FTE	Record configuration, event marker number, approach speed, RPM, MAP/power, and weight (SW121).
4	Call V_{50}	FTP	Maintain 60 KIAS and <u>announce airspeed</u> over obstacle.
5	Record data	FTE	Mark event + Chronometer START. Record GS and visual reference point.
6	Call touchdown	FTP	Maintain aligned heading; call out touchdown speed.
7	Record data	FTE	Mark event + Chronometer LAP. Log GS and visual reference.
8	Full stop	FTP	Complete braking and stop the aircraft safely.
9	Record data	FTE	Mark event + Chronometer LAP. Record stop point using visual reference.
10	Log conditions	FTE	Record OAT, QNH, QFE, weight, wind head and speed.

3 Safety considerations

Takeoff Testing

The take-off phase is inherently critical due to the short reaction time available in case of anomalies. The following table summarizes the main potential hazards identified during the test activity, along with corresponding mitigation or minimization strategies. For detailed risk levels, refer to the specific Test Hazard Analysis (THA) provided in Appendix A.

Hazard	Description	Mitigation
Engine Failure	Potential engine power loss or anomalies before rotation.	Perform thorough pre-flight engine checks (oil pressure/temp, RPM trends). Define and brief abort point. Apply brakes and aerodynamic deceleration immediately.
Brake Failure During Abort	Inability to safely decelerate during a rejected take-off.	Conduct brake check during taxi. Make sure to have sufficient length and a clear overrun zone to ensure safe stop.
Crosswind Exceedance	Risk of losing directional control due to strong lateral winds.	Take off only if crosswind is within aircraft limits. Use up-to-date weather briefing and confirm wind conditions before takeoff.
Runway Conditions	Reduced friction and possible loss of control on contaminated surfaces.	Avoid operations on wet, icy, or contaminated runways. Inspect and verify runway condition prior to the test.
Pilot Workload	Delayed control input nature of the test may lead to loss of situational awareness or delayed response in case of emergency.	Maintain high situational awareness. Ensure readiness to regain control promptly in case of an emergency. Follow recovery procedures as trained.

All take-off tests will be performed only when meteorological, operational, and aircraft conditions are fully compliant with safety criteria. Test execution will be preceded by a complete crew briefing and checklist review. In case of any non-conformance, the test will be postponed or aborted.

Phugoid Testing

The phugoid test, while relatively benign in terms of dynamic response, requires careful adherence to safety protocols due to the altitude changes and potential long-period oscillations involved. The following general and specific safety considerations apply:

Hazard	Description	Mitigation
Adverse Weather	Wind, turbulence, or thermal activity can distort phugoid behaviour, affect data quality, and cause excessive airspeed or stall.	Perform testing only in acceptable visual conditions with low wind and turbulence. Avoid gusty or thermally active environments. Monitor weather continuously.
Insufficient Altitude	Altitude loss during oscillations may reduce recovery margins.	Conduct all tests at a minimum of 3'000 ft. Abort test immediately if abnormal aircraft behaviour is observed.

Airspace Conflict	Testing in congested airspace may lead to proximity with other traffic, increasing collision risk.	Select a non-congested or pre-designated test area. Ensure self-coordination and, when possible, include a safety observer or pilot onboard.
Aggressive Control Input	Excessive speed deviation may excite non-phugoid modes or exceed structural/stall limits.	Limit initial speed deviation to <10% of cruise, performing the procedure taking 10 seconds. Avoid overshooting speeds to remain safely between stall and VNE limits in flap 0 configuration. Presence of parachute (SW121). Always maintain sufficient altitude for recovery.
Pilot Workload	Delayed control input nature of the test may lead to loss of situational awareness or delayed response in case of emergency.	Maintain high situational awareness. Refrain from unnecessary input during oscillations. Ensure readiness to regain control promptly in case of an emergency. Follow recovery procedures as trained.

Stall Testing

The stalling speed test aims to determine the minimum steady flight speed at which the aircraft is controllable, as required by certification regulations. Although stall entry is expected, the test is designed to avoid deep stall conditions or loss of control. The following table outlines key hazards and corresponding mitigation procedures.

Hazard	Description	Mitigation
Inadvertent Loss of Control	Risk of entering an uncontrolled stall or spin due to low speed or improper control use.	Build-up to stall. Pilot preparedness. Maintain sufficient altitude during testing. Presence of ballistic parachute (SW121).
Insufficient Altitude	Possibility of ground impact if recovery is delayed or ineffective.	Maintain minimum altitude margin. Avoid tests near terrain or populated areas. Review recovery procedures before flight.
Pilot-Induced Oscillations	Overcontrol may destabilize the aircraft and hinder stall recovery.	Stall approached gradually and smoothly. Conducted by a trained, experienced test pilot familiar with stall dynamics.
Adverse Weather	Wind gusts or turbulence can affect stall behaviour and control.	Perform tests only in calm air. Set and respect wind limitations.
Configuration Errors	Incorrect flap or trim settings may alter stall characteristics or increase risk.	Use standardized configuration checklist. Confirm flap and trim settings before each run, verified by pilot and test engineer.

All stalling speed tests will be conducted in predefined configuration, with stable flight path and trimmed conditions before stall onset. The aircraft shall be within the certified flight envelope and mass CG limits. A detailed briefing and emergency procedure review shall precede the test sequence.

Landing Testing

Landing tests, despite being part of routine flight operations, require strict safety precautions due to the increased risk associated with repeated touchdown manoeuvres, low altitudes, and potential runway excursions. The following considerations apply:

Hazard	Description	Mitigation
Adverse Weather	Wind, gusts, or poor visibility can compromise landing safety and data validity.	Conduct tests only in acceptable visual conditions with low wind. Avoid gusty or unstable atmospheric conditions. Continuously monitor meteorological updates.
Runway Conditions	Wet, short, or sloped runways can impair safe touchdown, braking, and rollout.	Use a dry, obstacle-free runway of adequate length. Document and evaluate runway slope and surface condition before testing.
Air Traffic Conflict	Shared use of runway with other traffic may cause timing issues or conflict.	Coordinate with ATZ to secure runway availability. Utilize a ground-based radio observer for safety communications and coordination.
Unsafe Touchdown	Excessive sink rate or abnormal touchdown might stress the airframe or invalidate the test.	Ensure acceptable descent rate. Abort test in case of bouncing or abnormal deceleration.
Brake Failure During Abort	Inability to safely decelerate during landing.	Conduct brake check during taxi. Make sure to have sufficient length and a clear overrun zone to ensure safe stop.
Pilot Workload	High workload or unclear responsibilities can reduce reaction time or compromise safety.	Keep pilot workload manageable. Conduct thorough pre-test briefing, including failure scenarios. Maintain clear task delegation and reinforce readiness for go-around or emergency response.

4 References

- [1] Gorizia Airport, *Official Website*. Available at: [Aeroporto di Gorizia "Amedeo Duca d'Aosta"](#).
- [2] ENAV, *Aeronautical Charts – PADOVA VFR AREA*.
- [3] ENAV, *AIP Italia – Sezione ENR*. Available at: [informazioni aeronautiche | Enav](#).
- [4] ASTM, *F2245 – Standard Specification for Design and Performance of Light Sport Airplanes*, ASTM International.
- [5] EASA, *CS-23, Amd 4 – Certification Specifications for Normal, Utility, Aerobatic, and Commuter Aeroplanes*.
- [6] Pipistrel, *Pilot's Operating Handbook (POH) – SW121*.
- [7] Pipistrel, *Pilot's Operating Handbook (POH) – SW128*.
- [8] J.D. Kimberlin, *Flight Testing of Fixed-Wing Aircraft*, AIAA Education Series, American Institute of Aeronautics and Astronautics, 2003.
- [9] U.S. Navy and U.S. Air Force Test Pilot Schools, *Takeoff and Landing Flight Testing*.
- [10] EASA, *CS-23, Amd 5 – Certification Specifications for Normal, Utility, Aerobatic, and Commuter Aeroplanes*.
- [11] Udine Airport, *Official Website*. Available at: [Control Zone di Aviano - Aeronautical Information](#)
- [12] WeatherSpark, *Average Weather in May in Gorizia, Italy*. Available at: <https://it.weatherspark.com/m/75102/5/Condizioni-meteorologiche-medie-nel-mese-di-maggio-a-Gorizia-Italia>

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Appendix

A Test Hazard Analysis

The Test Hazard Analysis (THA) provides a structured evaluation of potential hazards associated with the planned flight test activities. It identifies key risks, assesses their severity and likelihood, and outlines mitigation strategies to ensure flight safety. A risk management matrix is included to support this assessment. The following reference categories are used to classify hazard levels:

- **Category 1 – High (1–6, 8):** Elimination or mitigation actions must be taken to reduce the risk.
- **Category 2 – Low (7, 9–20):** Risk is considered acceptable.

The results are presented in dedicated tables for each test, showing the hazard level for each identified risk consideration.

Description	Category	Hazard Severity
Catastrophic	I	Death or serious injury to the public.
Critical	II	Major property damage to the public, major safety-critical system damage or reduced capability, significant reduction in safety margins, or significant increase in crew workload.
Marginal	III	Minor injury to the public or minor safety-critical damage.
Negligible	IV	Not serious enough to cause injury to the public or safety-critical system damage.

Description	Level	Individual Item
Frequent	A	Likely to occur often in the life of an item, with a likelihood of occurrence greater than 10^{-2} in any one mission.
Probable	B	Will occur several times in the life of an item, with a likelihood of occurrence less than 10^{-2} but greater than 10^{-3} in any one mission.
Occasional	C	Likely to occur sometime in the life of an item, with a likelihood of occurrence less than 10^{-3} but greater than 10^{-5} in any one mission.
Remote	D	Unlikely but possible to occur in the life of an item, with a likelihood of occurrence less than 10^{-5} but greater than 10^{-6} in any one mission.
Extremely Remote	E	So unlikely, it can be assumed occurrence may not be experienced, with a likelihood of occurrence less than 10^{-6} in any one mission.

Risk hazard Index				
	I	II	III	IV
A	1	3	7	13
B	2	5	9	16
C	4	6	11	18
D	8	10	14	19
E	12	15	17	20

Test Hazard Analysis – TAKE-OFF

Hazard	Risk Index	Mitigation (if Risk ≤ 8)	Risk after mitig.
Engine failure	4	Pre-flight engine checks (oil pressure/temp, RPM trends). Define and brief abort point. Apply brakes and aerodynamic deceleration immediately.	10
Brake failure during abort	10	—	—
Crosswind exceedance	18	—	—
Runway conditions	14	—	—
Pilot Workload	11	Maintain high situational awareness. Ensure readiness in case of an emergency. Follow recovery procedures as trained.	—

Test Hazard Analysis – PHUGOID

Hazard	Risk Index	Mitigation (if Risk ≤ 8)	Risk after mitig.
Adverse weather	7	Perform testing only in acceptable visual conditions with low wind and turbulence. Avoid gusty or thermally active environments. Monitor weather continuously	—
Airspace conflict	18	—	—
Insufficient altitude	6	Conduct all tests at a minimum of 3'000 ft. Abort test immediately if abnormal aircraft behaviour is observed.	14
Aggressive control input	9	—	—
Pilot workload	18	—	—

Test Hazard Analysis – STALL

Hazard	Risk Index	Mitigation (if Risk ≤ 8)	Risk after mitig.
Inadvertent loss of control	5	Build-up to stall. Pilot preparedness. Maintain sufficient altitude during testing. Presence of ballistic parachute (SW121).	10
Insufficient altitude	8	Maintain minimum altitude margin. Avoid tests near terrain or populated areas. Review recovery procedures before flight.	12
Pilot-induced oscillation	18	—	—
Weather conditions	13	—	—
Configuration error	19	—	—

Test Hazard Analysis – LANDING

Hazard	Risk Index	Mitigation (if Risk ≤ 8)	Risk after mitig.
Adverse weather	7	Conduct tests only in acceptable visual conditions with low wind. Avoid gusty or unstable atmospheric conditions. Continuously monitor meteorological updates.	—
Runway conditions	11	—	—
Air traffic	16	—	—
Unsafe touchdown	6	Ensure acceptable descent rate. Abort test in case of bouncing or abnormal deceleration.	14
Brake failure	10	—	—
Pilot workload	10	—	—